

**Corrections to Preparatory Problems for
the 58th International Chemistry Olympiad 2026
Tashkent, Uzbekistan**

Theoretical Problems

Problem 8. Step-by-step

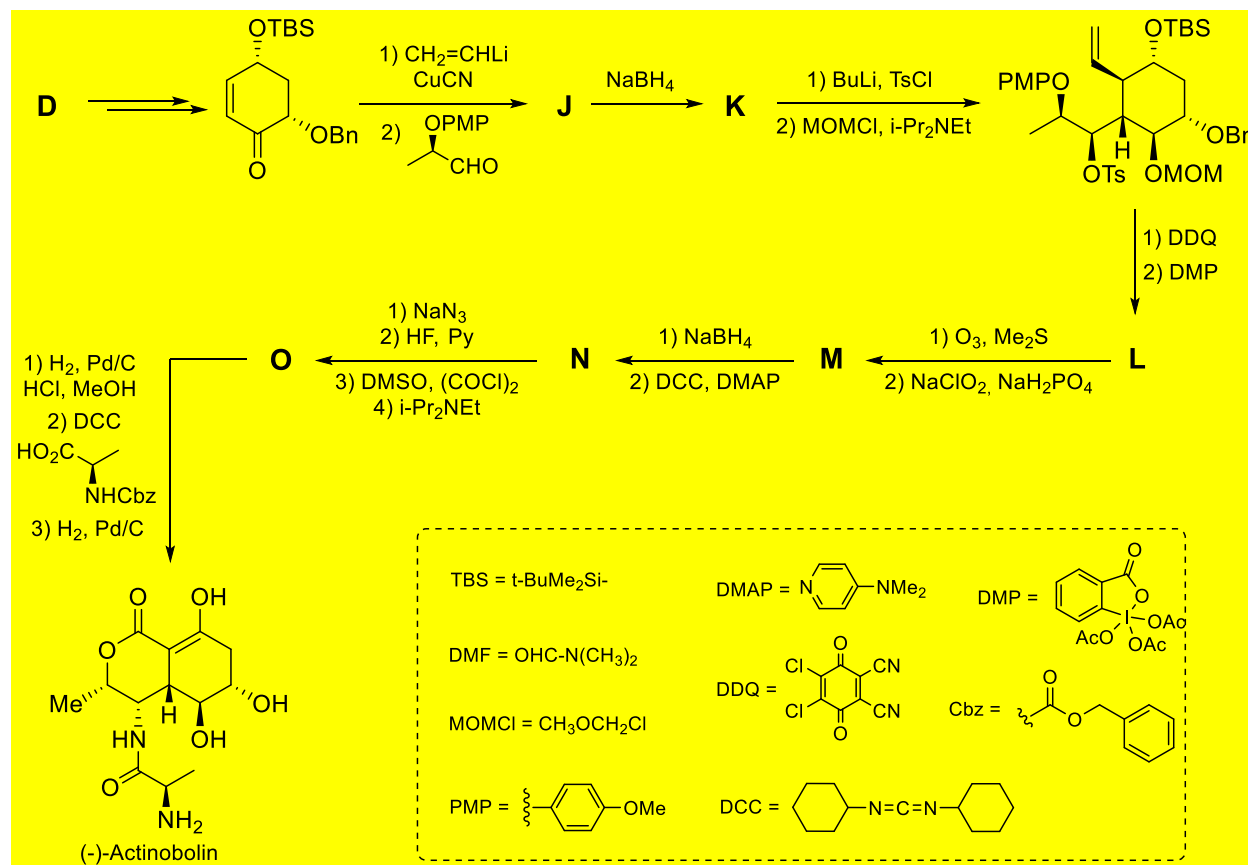
10. **Derive** the relationship $DP = f(t)$ for $\xi > 1$ by taking into account the following assumptions (even though they may not be true in experimental polycondensation reaction conditions):

- The concentration of reactive chain ends remains constant.
- All polymer molecules possess identical chain lengths.
- Each propagation step adds only a single monomer to a chain.

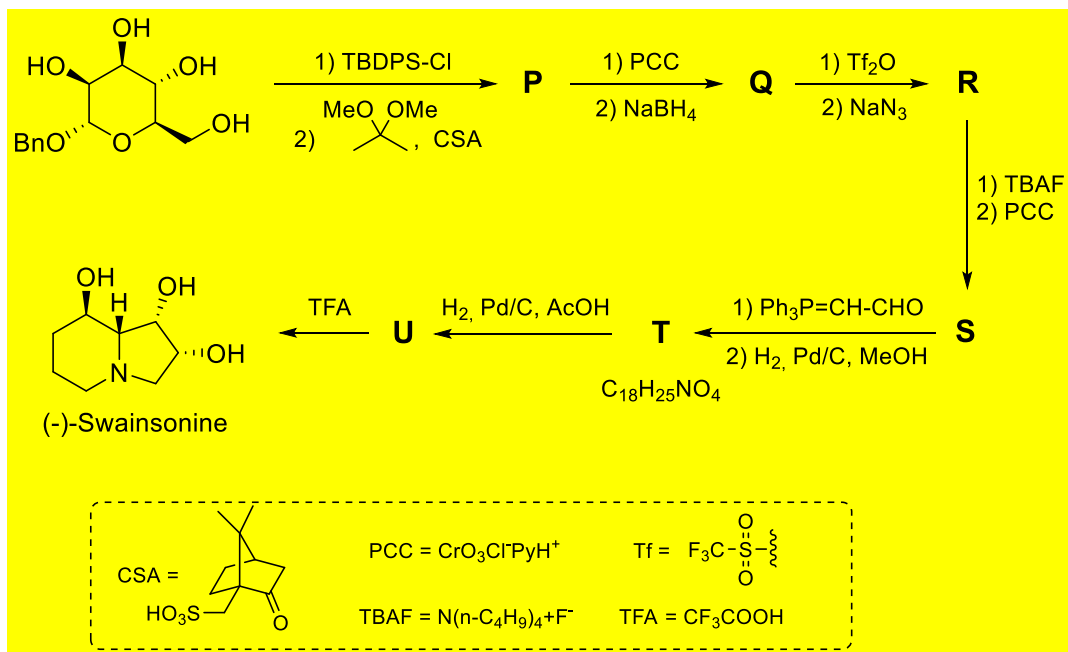
Problem 19. Carbohydrates as chiral pools

Asymmetric synthesis has attracted significant attention in recent years thanks to advances in asymmetric catalysis. Nevertheless, the transformation of stereocenters in readily available chiral substrates remains an efficient approach to enantioselective total synthesis, commonly referred to as the “chiral pool” strategy. Various monosaccharides serve as convenient starting materials in natural product synthesis. For example, in 1985, the first synthesis of compound **X** from L-sorbose derivative **A** was published.

In the synthesis of (+)-Actinobolin, overall, five new stereocentres were formed using the chiral auxiliary.



The synthesis of (–)-Swainsonine is given below.



Problem 30. Pilaf ingredients

Upon oxidative ozonolysis, acyclic compound **F**, without any triple bond, having one oxygen-containing group, yields an equimolar mixture of $\text{C}_\alpha\text{H}_\beta\text{O}_\gamma$ (**F**₁) and $\text{C}_\delta\text{H}_\epsilon\text{O}_\alpha$ (**F**₂), where $\alpha \neq \beta \neq \gamma \neq \delta \neq \epsilon$.